

A shrinking-skeleton positive criterion for fully Mackey complete duals

Run `fa_banach_001`, agent `agent_lane_05`

June 14, 2026

Status

This packet records a claimed partial positive answer to Problem 4.15 of Guirao, Martinez-Cervantes and Rodriguez [1]. It is not a full solution of the problem. It proves the desired implication for Banach spaces whose bidual is controlled by a shrinking separable projectional skeleton.

Source problem

Guirao, Martinez-Cervantes and Rodriguez [1, Problem 4.15] ask:

Is X^* fully Mackey complete whenever X is w^* -sequentially dense in X^{**} ?

The relevant source characterization is [1, Theorem 3.5]: a Banach space W is fully Mackey complete if and only if it satisfies property (LK) : for every norming norm-closed subspace $Y \subset W^*$ and every $u \in W^* \setminus Y$, there is an absolutely convex weak-star compact $K \subset W^*$ such that

$$K \subset Y \oplus [u], \quad u \in \overline{K \cap Y}^{w^*}.$$

14

A.J. GUIRAO, G. MARTÍNEZ-CERVANTES, AND J. RODRÍGUEZ

Problem 4.15. *Is X^* fully Mackey complete whenever X is w^* -sequentially dense in X^{**} ?*

Remark 4.7 makes clear that a negative answer to Problem 4.15 would be based

Proof Intuition

The previous ℓ_p -sum packet showed that countable coordinate supports let one localize a norming subspace $Y \subset X^{**}$ to a separable countable sub-sum. The present packet isolates the abstract mechanism behind that argument.

Suppose X has many separable projection ranges $P_s X$, every point of X^{**} is fixed by some P_s^{**} , and the projections are shrinking along countable suprema. Given a norming closed $Y \subset X^{**}$ and a starting range containing u , recursively do the following. On the current separable range, the dual ball is weak-star compact metrizable. A countable subcover therefore selects countably many

elements of Y which norm that current range. Since every selected element of Y is captured somewhere by the skeleton, move to a later range containing all of them. At the countable supremum, the selected elements lie in Y intersected with the final bidual range, and the shrinking hypothesis passes the norming estimates to the whole final dual.

The source paper's separable theorem then applies to the final separable range: it contains no ℓ_1 , so its dual is fully Mackey complete. Its LK witness embeds back into X^{**} because the final range is the range of a projection.

Definitions

Let $(\Sigma, \leq)$ be a directed partially ordered set in which every increasing sequence has a supremum.

Definition 1. A family $(P_s)_{s \in \Sigma}$ of bounded projections on a Banach space X is called a *shrinking bidual-capturing separable projectional skeleton* if:

1. $\sup_s \|P_s\| < \infty$ and $P_s X$ is separable for every s ;
2. if $s \leq t$, then $P_s P_t = P_t P_s = P_s$;
3. if $s = \sup_n s_n$ for an increasing sequence (s_n) , then $P_s X = \overline{\bigcup_n P_{s_n} X}$ and, for every $\phi \in (P_s X)^*$,
$$\|\phi - \phi \circ P_{s_n}|_{P_s X}\|_{(P_s X)^*} \rightarrow 0;$$
4. for every $x^{**} \in X^{**}$ there is $s \in \Sigma$ such that $P_s^{**} x^{**} = x^{**}$.

Main theorem

Theorem 1. Let X be a Banach space containing no subspace isomorphic to ℓ_1 . Assume that X admits a shrinking bidual-capturing separable projectional skeleton. Then X is weak-star sequentially dense in X^{**} , and X^* is fully Mackey complete.

Localization lemma

For $s \in \Sigma$, put $E_s = P_s X$ and $R_s = P_s^{**} X^{**}$. We identify R_s with E_s^{**} in the usual way. The weak-star topology induced from X^{**} on R_s agrees with $\sigma(E_s^{**}, E_s^*)$, since every member of E_s^* extends to X^* .

Lemma 1. Let $Y \subset X^{**}$ be a norming norm-closed subspace. For every $s_0 \in \Sigma$ there is $s \geq s_0$ such that

$$Y_s := Y \cap R_s$$

is norming for E_s^* .

Proof. Let $M = \sup_s \|P_s\|$. Choose $0 < \delta \leq 1$ such that

$$\sup\{|y(x^*)| : y \in Y, \|y\| \leq 1\} \geq \delta \|x^*\| \quad (x^* \in X^*). \quad (1)$$

We recursively construct an increasing sequence $s_0 \leq s_1 \leq s_2 \leq \dots$ and countable sets $C_n \subset Y \cap B_{X^{**}}$. Suppose s_n has been chosen. Since E_{s_n} is separable, the compact space $(B_{E_{s_n}^*}, w^*)$ is metrizable. For each rational $r \in (0, 1)$, set

$$O_r = \{\phi \in B_{E_{s_n}^*} : \|\phi\| > r\}.$$

For $\phi \in O_r$, apply (1) to $P_{s_n}^* \phi = \phi \circ P_{s_n} \in X^*$. Because $P_{s_n}^* \phi$ restricts to ϕ on E_{s_n} , we have $\|P_{s_n}^* \phi\| \geq \|\phi\| > r$. Hence some $y \in Y \cap B_{X^{**}}$ satisfies

$$|P_{s_n}^{**} y(\phi)| = |y(P_{s_n}^* \phi)| > \delta r.$$

Thus the weak-star open sets

$$\{\psi \in B_{E_{s_n}^*} : |P_{s_n}^{**} y(\psi)| > \delta r/2\}, \quad y \in Y \cap B_{X^{**}},$$

cover O_r . Since O_r is Lindelof, choose a countable subcover. Taking the union over rational $r \in (0, 1)$ gives a countable set $C_n \subset Y \cap B_{X^{**}}$ such that

$$\sup_{y \in C_n} |P_{s_n}^{**} y(\phi)| \geq \frac{\delta}{2} \|\phi\| \quad (\phi \in E_{s_n}^*). \quad (2)$$

For every $y \in C_n$, choose $t_y \in \Sigma$ such that $P_{t_y}^{**} y = y$. Using directedness and countable suprema, choose $s_{n+1} \geq s_n$ which dominates all t_y with $y \in C_n$. Then $C_n \subset Y \cap R_{s_{n+1}}$.

Let $s = \sup_n s_n$. Then $C_n \subset Y \cap R_s$ for every n . We claim that Y_s is norming for E_s^* . Let $0 \neq \phi \in E_s^*$, and write

$$\phi_n := \phi \circ P_{s_n}|_{E_s}.$$

By the shrinking assumption, $\|\phi - \phi_n\| \rightarrow 0$. Also $\phi_n|_{E_{s_n}} \in E_{s_n}^*$, and

$$\|\phi_n\| \leq M \|\phi_n|_{E_{s_n}}\|.$$

Fix a small $\eta > 0$ and choose n so large that

$$\|\phi - \phi_n\| < \eta \|\phi\|, \quad \|\phi_n|_{E_{s_n}}\| > \frac{1-\eta}{M} \|\phi\|.$$

By (2), choose $y \in C_n$ within $\eta \|\phi\|$ of the supremum at $\phi_n|_{E_{s_n}}$. Since $y \in R_s$ and $\|y\| \leq 1$,

$$\begin{aligned} |y(\phi)| &\geq |y(\phi_n)| - |y(\phi - \phi_n)| \\ &= |P_{s_n}^{**} y(\phi_n|_{E_{s_n}})| - |y(\phi - \phi_n)| \\ &> \left(\frac{\delta}{2M} (1-\eta) - 2\eta \right) \|\phi\|. \end{aligned}$$

Taking η sufficiently small gives

$$\sup\{|y(\phi)| : y \in Y_s, \|y\| \leq 1\} \geq \frac{\delta}{4M} \|\phi\|.$$

Therefore Y_s is norming for E_s^* . □

Proof of Theorem 1

First let $u \in X^{**}$. Choose s with $P_s^{**} u = u$. The range E_s is separable and contains no ℓ_1 , because it is a subspace of X . By [1, Corollary 4.8], $(B_{E_s^{**}}, w^*)$ is Frechet-Urysohn. Since E_s is weak-star dense in E_s^{**} by Goldstine's theorem, there is a sequence in $E_s \subset X$ converging weak-star to u . Hence X is weak-star sequentially dense in X^{**} .

It remains to show that X^* is fully Mackey complete. We verify property (LK) from [1, Theorem 3.5]. Let $Y \subset X^{**}$ be norming and norm-closed, and let $u \in X^{**} \setminus Y$. Choose s_0 with $P_{s_0}^{**} u = u$.

By Lemma 1, after increasing s_0 if necessary, we may assume that $Y_s = Y \cap R_s$ is norming for E_s^* and still $u \in R_s$.

The space E_s is separable and contains no ℓ_1 , so [1, Corollary 4.8] gives that E_s^* is fully Mackey complete. Applying property (LK) in the dual space E_s^* to the norming closed subspace $Y_s \subset E_s^{**}$ and to $u \notin Y_s$, we obtain an absolutely convex weak-star compact set $K \subset R_s$ such that

$$K \subset Y_s \oplus [u], \quad u \in \overline{K \cap Y_s}^{\sigma(E_s^{**}, E_s^*)}.$$

Regarded as a subset of X^{**} , the same K is weak-star compact, because the weak-star topology on R_s is the one induced from X^{**} . Moreover $Y_s \subset Y$, and therefore

$$K \subset Y \oplus [u], \quad u \in \overline{K \cap Y}^{\sigma(X^{**}, X^*)}.$$

This is the required LK witness. Thus X^* is fully Mackey complete.

Relation to the earlier ℓ_p -sum packet

The criterion recovers the previous positive theorem for

$$X = \left(\bigoplus_{\gamma \in \Gamma} E_\gamma \right)_{\ell_p}, \quad 1 < p < \infty,$$

when each E_γ is separable and contains no ℓ_1 . Take Σ to be the countable subsets of Γ and let P_A be the coordinate projection. Every element of $X^{**} = (\bigoplus_\gamma E_\gamma^{**})_{\ell_p}$ has countable support, so it is captured by some P_A^{**} . Along increasing countable sets, the dual tails in $X^* = (\bigoplus_\gamma E_\gamma^*)_{\ell_q}$ converge in norm. Finally, X contains no ℓ_1 : any bounded sequence is supported in a countable sub-sum, and the standard Rosenthal diagonal argument applies there.

Verification and limitations

This is a partial positive result, not a full solution of Problem 4.15. It identifies a sufficient structural condition under which the countable localization argument works. The proof does not show that every Banach space which is weak-star sequentially dense in its bidual has such a skeleton. In fact, that is exactly the obstruction exposed by the argument: weak-star sequential density supplies separable weak-star closures of individual bidual points, but it does not by itself supply projections whose adjoints converge in norm on the relevant separable duals.

The main point for review is Lemma 1, especially the last passage from the estimates on $E_{s_n}^*$ to the estimate on E_s^* . This is where the shrinking condition is used and where a full solution would need a replacement mechanism.

References

- [1] A. J. Guirao, G. Martinez-Cervantes, and J. Rodriguez, *Completeness in the Mackey topology by norming subspaces*, J. Math. Anal. Appl. **478** (2019), no. 2, 776–789; arXiv:1812.10669.